

Chapter 6 - Practice Set

Tuples

Write each program below as its own separate Python file (for example, q1.py, q2.py, and so on) and run it to check your work.

Question 1

Write a program that stores a tuple of six of your favorite songs and prints: the full tuple, the first song, the last song, the total number of songs using `len()`, and a slice containing only the second and third songs.

Question 2

Write a program that creates two variables, `a = (5)` and `b = (5,)`, and prints the type of each one using `type()`, to demonstrate the difference between a plain value in parentheses and a true single-element tuple.

Question 3

Write a program that stores the tuple `("Mon", "Tue", "Wed")` in a variable and attempts to change the first element to `"Sunday"` using index assignment, so that Python raises an error. Below that (commented out or explained with a print statement), show the correct way to build a new tuple with `"Sunday"` as the first element instead.

Question 4

Write a program that stores a tuple containing a person's name, age, and city, then unpacks all three values into three separate variables in a single line and prints each variable on its own line.

Question 5

Write a program that stores a tuple of seven numbers and uses starred unpacking (for example `a, *b, c = numbers`) to assign the first number, the last number, and all the numbers in between to three variables. Print all three variables.

Question 6

Write a program that stores a tuple with at least one repeated value and uses `count()` to print how many times a specific value appears, and `index()` to print the position of the first occurrence of a different value.

Question 7

Write a program that stores a tuple of valid product codes and asks the user to type a code using `input()`. Using the `in` and `not in` operators, print a message stating whether or not the typed code is a valid product code.

Question 8

Write a program that creates a nested tuple representing three students, where each inner tuple contains a student's name and their score, for example `("Maya", 88)`. Print the entire nested tuple, then use double indexing to print just the name of the second student and just the score of the third student.

Question 9

Write a program that stores a list of daily temperatures that will keep changing throughout the day, and a tuple of fixed (latitude, longitude) coordinates for a weather station that should never change. Demonstrate that the list can be modified with `append()`, and add a comment explaining why a tuple was chosen for the coordinates instead of a list.

Question 10

Write a program that stores a tuple representing a fixed RGB color value, for example `(255, 0, 0)`. Print each of the three color channels separately by indexing into the tuple, and print a message confirming the total number of channels using `len()`.